

## OLAP, OLTP, and NewSQL Database Systems

In this train, we look at OLAP and OLTP databases and why they are important . We also present NewSQL , a new database type, and discuss how it differs from relational and NoSQL databases.

### Learning Objectives

By the end of this train you should be able to:

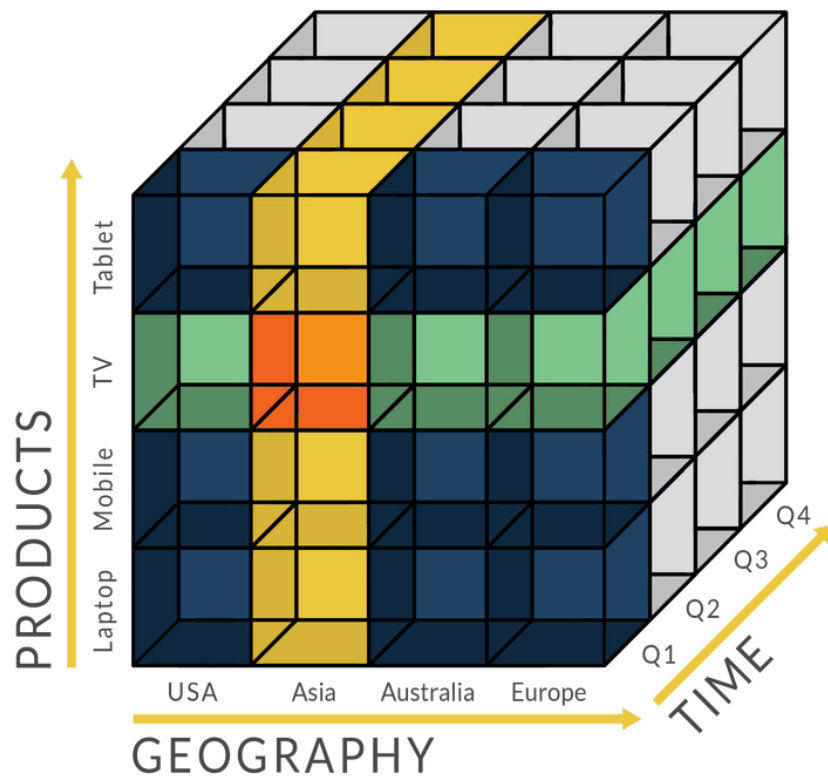
- Differentiate between OLAP, OLTP, and NewSQL databases, understanding their core features, use cases, and operational differences.
- Explore how NewSQL combines the scalability of NoSQL with the consistency of traditional relational databases to meet modern data management needs.

### OLAP and OLTP Databases

The increase in the amount, type, and sources of data over the past three decades has led to the birth of two distinct database categories: OLAP and OLTP database. These categories are suited for analytical, online analytical processing (OLAP), and transactional, online transaction processing (OLTP) workloads.

#### OLAP Databases

Analysts frequently need to group, aggregate, and join data. These operations in relational databases are resource-intensive. With OLAP, data can be pre-calculated and pre-aggregated, making analysis faster. OLAP databases are divided into one or more OLAP cubes. The cubes are designed in such a way that creating and viewing reports becomes easy.



*Figure 9: An OLAP cube illustration. A three-dimensional array is built (the cube) to easily query specific product sales by time of year for a specific geographic region.*

These cubes can then be navigated using four basic types of analytical operations in OLAP databases:

1. **Roll-up:** View in decreased detail.
2. **Drill-down:** View in increased detail.
3. **Slice and dice:** Focuses on one particular dimension of a cube and provides a new sub-cube to analyze. For example, analyzing only the Asian "sub-cube" for weekly TV sales data in Q1.
4. **Pivot (rotate):** Rotate the data to provide an alternative representation by switching the x, y, and z axes of the cube.

Usually, the number of users querying an OLAP database is relatively low, primarily consisting of the employees of a company only.

## OLTP Databases

OLTP databases are concerned with all transactions, including short-lived, pre-defined queries. OLTP databases are designed with less detail in mind than OLAP databases but need to be built for a much larger number of users.

## Examples of OLTP transactions

- Search results on an e-commerce website.
- Online banking.
- Online airline ticket booking.
- Sending text messages.
- Grocery purchases via an app or website.
- Purchasing a product on an e-commerce website.

OLTP typically involves:

### Inserting

The creation of new records as part of high-volume, real-time transactional operations. These insertions are typically small and involve a minimal number of rows or entries.

### Updating

Modifying existing records in response to real-time transactional changes. This operation ensures that the database reflects the most accurate and current state of the data

### Deleting

Removing specific records from a database as part of a transactional workflow. These operations are typically small-scale and executed in real-time to keep the database clean and performant.

When designing an OLTP system, it is important to ensure the following:

- Short response times.
- Only small transformations are applied.
- The system has high concurrency.
- The system can handle large volumes of continuous data streams.
- The system is highly available.

**Comparison of OLAP and OLTP Database Systems:**

| Properties        | OLAP                         | OLTP                                                     |
|-------------------|------------------------------|----------------------------------------------------------|
| Use case          | Used for data analysis       | Manages a very large number of online short transactions |
| Database type     | Uses data warehouses         | Uses traditional DBMSs                                   |
| Data modification | Mainly used for data reading | Manages all insert, update, and delete transactions      |
| Response time     | Processing is a little slow  | Processes in milliseconds                                |
| Normalization     | Tables are not normalized    | Tables are normalized                                    |

## The New Player, NewSQL

NoSQL is very useful for applications such as social media, where eventual consistency is acceptable. This means that users rarely notice or have their user experience ruined if they see a non-consistent view of the database. However, NoSQL databases are not easy to use for systems where consistency is critical, such as banking and finance platforms. This is where NewSQL comes into play.

NewSQL systems combine the scalability and high availability of NoSQL with the relational model, transaction support, and SQL of traditional relational databases. NewSQL databases are built for the cloud era, with a horizontally scalable architecture in mind from the start.

Such systems aim to achieve the scalability of NoSQL systems while still providing the ACID attributes of traditional relational databases. They are ideal for companies handling high-profile data requiring scalability and consistency, such as financial institutions.

Feature comparison:

| Feature               | SQL (Relational Databases) | NoSQL                    | NewSQL                                                                 |
|-----------------------|----------------------------|--------------------------|------------------------------------------------------------------------|
| Relational property   | Yes                        | No                       | Yes                                                                    |
| ACID                  | Yes                        | No                       | Yes                                                                    |
| SQL                   | Yes                        | No                       | Yes, with enhanced performance                                         |
| OLTP                  | Full                       | Limited                  | Full                                                                   |
| Scaling               | Vertical                   | Vertical and horizontal  | Vertical and horizontal                                                |
| Query handling        | Best for simple queries    | Best for complex queries | Improved performance for simple queries and capable of complex queries |
| Distributed databases | No                         | Yes                      | Yes                                                                    |

## Identifying the Right Database Type

Scenario: Imagine you are a database engineer at a growing fintech company. Your tasks include:

1. Ensuring real-time transaction processing for customer payments.
2. Generating weekly reports that analyze customer spending trends across regions.
3. Scaling the system to accommodate a growing customer base without compromising on data consistency.

## Conclusion

In this lesson, we explored the differences between OLAP, OLTP, and NewSQL database systems, understanding their unique features and use cases. While OLAP excels in handling analytical workloads and aggregating large datasets, OLTP is designed for real-time transactional processing. The emergence of NewSQL bridges the gap by combining the scalability of NoSQL with the consistency and relational properties of traditional databases.

### Key Takeaways

- **Takeaway 1: OLAP (Online Analytical Processing)** Designed for complex analytical queries and aggregations on historical data, typically using data warehouses and OLAP cubes.
- **Takeaway 2: OLTP (Online Transaction Processing)** Optimized for high-volume, short, real-time transactions (e.g., inserts, updates, deletes) with a focus on quick response times and high concurrency.
- **Takeaway 3: NewSQL Databases** A hybrid approach combining the horizontal scalability and high availability of NoSQL with the relational model, ACID compliance, and SQL support of traditional relational databases.
- **Takeaway 4: Purpose-Built Systems** OLAP and OLTP serve distinct purposes (analytics vs. transactions), requiring different database designs and optimizations.
- **Takeaway 5: NewSQL's Role** Bridges the gap for applications needing both relational consistency and NoSQL-like scalability, common in modern high-performance transactional systems.